Albin Zeqiri

Ulm, Germany zeqiri.albin@outlook.de Google Scholar LinkedIn GitHub Website

Human-Computer Interaction Researcher/Doctoral Candidate

I am a **final-year PhD candidate in Human–Computer Interaction** at Ulm University, with a research focus on sustainability and carbon reduction in resource-intensive digital systems. My work examines how carbon-reduction mechanisms influence subsequent user behavior and how emerging technologies (e.g., civic robotics) can be implemented to support just resource distribution. Methodologically, I combine quantitative and qualitative approaches. I have a background in HCI (BSc), Data Science (BSc, MSc), and Behavioral Psychology, and have published at top-tier HCI venues, including **CHI** and **IMWUT**. I regularly collaborate with academic partners (**UCL Interaction Centre, Karlsruhe Institute of Technology**) and industry partners (**Mercedes-Benz Tech Innovation**).

Skills

Experimental Design: Online, laboratory, and field-based experiments (exploratory and hypothesis-driven); large-scale literature surveys (e.g., PRISMA); data mining using multi-API data collection and NLP-based analysis
Analysis: Statistical analysis (frequentist and Bayesian); machine learning and deep learning (supervised and self-supervised), qualitative analysis using thematic analysis and grounded theory
Programming Languages: Python, R, C#, Java, C/C++, SQL, Bash, HTML, CSS, JavaScript
Frameworks and Tools: PyTorch, TensorFlow, Keras, ROS, CUDA, OpenCV, Git, Gradle, JUnit, MySQL, PostgreSQL
Design: UI/UX prototyping and wireframing; image and video editing (Adobe Creative Cloud)
Languages: Albanian (native), German (native), English (fluent), French (intermediate)

Employment

ULM UNIVERSITY - Chair of Human-Computer Interaction Ulm, Germany
Research Associate 09/2022 – present

- Led HCI research on carbon-reduction mechanisms in resource-intensive systems across residential, automotive, and online contexts, integrating quantitative and qualitative evaluation methods.
- Planned and executed 10+ research studies, including controlled experiments, in-the-wild deployments, and work focused specifically on data mining/dataset curation as input for AI-based systems
- Collaborated with academic (**UCL Interaction Centre**, **Karlsruhe Institute of Technology**) and industry research partners (**Mercedes-Benz Tech Innovation**), with primary responsibility for designing reproducible deep learning pipelines and ablation studies, leading to two peer-reviewed publications.
- Supervised and mentored 40+ undergraduate and graduate students, providing guidance through problem formulation, study design, data analysis, and publication.

ULM UNIVERSITY – Chair of Human–Computer Interaction/Visual Computing Ulm, Germany
Research Assistant/Tutoring 12/2019 – 04/2022

- Developed interactive VR research prototypes using Unity and C#
- Supported the design, execution, and evaluation of empirical user studies
- Taught C#, Unity, and fundamentals of UI/UX prototyping to a class of 40+ students
- Managed and maintained course materials for supervising faculty
- Evaluation and reporting of various monocular depth estimation models in terms of performance and scalability

Education

Ulm University, Ulm, Germany 09/2022 – 12/2026 (expected)
PhD Candidate in Human-Computer Interaction
Dissertation Working Title: Carbon Reduction Mechanisms in Resource-Intensive Digital Systems: A Sufficiency-Based Approach to User-Centered Design
Research Areas: Responsible and Sustainability Computing, Design Tradeoff Optimization
Advisor: Prof. Dr. Enrico Rukzio

Ulm University, Ulm, Germany 03/2020 – 08/2022
M.Sc. Computer Science Overall Grade: 1.3 (A-equivalent)
Thesis Title: A Dataset and Temporal Modeling Approach for Automated Thermal Comfort State Recognition Thesis Grade: 1.0 (A-equivalent)
Published at UbiComp '24: 10.1145/3678503

Ulm University, Ulm, Germany 10/2016 – 12/2019
B.Sc. Computer Science Overall Grade: 2.0 (B-equivalent)
Thesis Title: Depth Levels: Measuring Achievable Levels of Voluntary Vergence Eye Movements for Eye-based Human-Computer Interaction Thesis Grade: 1.0 (A-equivalent)

Publications (Excerpt)

A. Zeqiri, M. Rietzler, and E. Rukzio. 2026. Investigating the Effects of Eco-Friendly Service Options on Rebound Behavior in Ride-Hailing. In Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems (CHI '26). 10.1145/3772318.3790711

M. Sasalovici, **A. Zeqiri**, R. C. Schramm, O. J. A. Nunez, P. Jansen, J. P. Freiwald, M. Colley, C. Winkler, and E. Rukzio. 2025. *Bumpy Ride? Understanding the Effects of External Forces on Spatial Interactions in Moving Vehicles.* In Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25). 10.1145/3706598.3714077.

A. Zeqiri, J. Britten, C. Schramm, P. Jansen, M. Rietzler, and E. Rukzio. 2025. *PlantPal: Leveraging Precision Agriculture Robots to Facilitate Remote Engagement in Urban Gardening.* In Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25). 10.1145/3706598.3713180.

A. Zeqiri, P. Jansen, J. O. Rixen, M. Rietzler, and E. Rukzio. 2024. *'Eco Is Just Marketing': Unraveling Everyday Barriers to the Adoption of Energy-Saving Features in Major Home Appliances.* In Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT). 10.1145/3643558.

M. Colley*, S. Hartwig*, **A. Zeqiri**, T. Ropinski, and E. Rukzio. 2024. *AutoTherm: A Dataset and Benchmark for Thermal Comfort Estimation Indoors and in Vehicles.* Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT). 10.1145/3678503.

Other

Secured competitive funding for my own doctoral research and collaborative lab projects.

- Received the *Landesgraduiertenförderung (LGFG)* doctoral scholarship, a competitive three-year fellowship supporting PhD research in Germany
- Received a one-time travel grant (€2800) from the *Graduate & Professional Training Center Ulm*
- Supported a successful Deutsche Forschungsgemeinschaft (DFG) Reinhart Koselleck proposal, *VRooms: Fighting Climate Change by Increasing the Utilization of Buildings through Everyday Extended Reality* (project link), with Prof. Dr. Enrico Rukzio